# GYANA RANJAN SAHOO

+91-8480882769 | gyanaranjansahoo174@gmail.com | LinkedIn | GitHub | Portfolio
LeetCode | GeeksForGeeks | Coding Ninjas

## Professional Summary

Computer Engineering student (B.Tech, graduating 2027) with a strong foundation in **Data Structures & Algorithms**, **OOP**, **DBMS**, **Operating Systems** and **Computer Networks**. Experienced in building **REST APIs** and **microservices** with Node.js/Express and TypeScript, backed by **SQL** databases (PostgreSQL, MySQL), with **Java** as a core language. Built an AI-powered platform that streams **LLM-generated code** into a live preview, and certified in **Generative AI Systems**. Comfortable with **Git**, **Docker**, **AWS** and CI/CD; active open-source contributor and competitive problem solver.

## Education

**Siksha 'O' Anusandhan (ITER)** — Bhubaneswar, India
*Bachelor of Technology in Computer Engineering (CGPA: 7.81)* — *Sept 2023 – Expected July 2027*

## Technical Skills

**Languages:** Java, SQL, C++, TypeScript, JavaScript (ES6+), C, Python (Working Knowledge)
**Backend & APIs:** RESTful API Design, Microservices, Node.js, Express.js, JWT Authentication, WebSockets
**Databases:** PostgreSQL, MySQL, SQL Query Optimization, Relational Schema Design, Prisma ORM, Drizzle ORM
**Core CS Foundations:** Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks
**DevOps & Cloud:** Git, Docker, Linux, AWS (EC2, S3, Auto Scaling), CI/CD (GitHub Actions), Prometheus, Grafana
**Generative AI & AI Tools:** Claude Code (AI coding assistant), LLM-based code generation workflows (project experience), Generative AI Systems (Microsoft certified)
**Frontend & Mobile:** ReactJS, React Native, Redux, HTML5, CSS3, Tailwind CSS

## Projects

**App Forge — AI Mobile App Builder** | *Next.js, React Native, TypeScript, Prisma, AWS, Docker* — GitHub | Live
- Engineered an AI-powered platform that transforms natural-language prompts into production-ready React Native (Expo) applications, streaming **LLM-generated code** into a live in-browser WebContainer preview.
- Architected a scalable Bun + Turborepo monorepo of four **microservices** (Next.js frontend, Express API, generation worker, AWS EC2 Auto Scaling orchestrator) secured with Clerk auth and a Prisma/PostgreSQL data layer.
- Containerized every service with **Docker**, automated multi-image **CI/CD** via GitHub Actions, and deployed across Vercel, Render, and Neon with a server-side Expo tunnel for instant on-device previews.

**Sonexa — Social Music Streaming Platform** | *Next.js, TypeScript, Prisma, PostgreSQL, Docker* — GitHub | Live
- Built a full-stack web application with instant track search, YouTube/Spotify playback, and a creator dashboard for managing live queues.
- Implemented real-time listening rooms with synchronized playback, presence tracking, and community upvoting to shape the shared queue.
- Architected a pnpm + Turborepo monorepo with a shared **Prisma** DB package, containerized deployment, and GitHub Actions **CI/CD** automation for Docker builds.

**Collaborative Real-Time Whiteboard** | *Next.js, TypeScript, WebSockets, Prisma, Docker* — GitHub | Live
- Engineered a high-fidelity, touch-responsive collaborative canvas using the Raw HTML5 Canvas API, optimizing client-side performance and frame rendering for seamless cross-device compatibility.
- Architected a custom **WebSocket protocol** to handle asynchronous events, ensuring state synchronization across clients with sub-50ms latency.

**Freshly — Cross-Platform Grocery & Meal Planner** | *React Native, Expo, TypeScript, PostgreSQL* — GitHub | Live
- Built a cross-platform (iOS, Android, Web) grocery list and meal planner app with Expo Router file-based routing and typed API routes, backed by Clerk authentication and secure token storage.
- Designed a relational **PostgreSQL** schema via Drizzle ORM on Neon's serverless driver, exposing **REST** API endpoints for item CRUD, bulk-clear operations, and category/spending insights.
- Automated EAS builds/OTA updates and lint/typecheck **CI/CD** via GitHub Actions, with Sentry error tracking in production; implemented global state with Zustand and a NativeWind/Tailwind UI.

## Certifications & Awards

**KFintech Internship — Applied Fintech Engineering & Mutual Fund Technology:** Completed program by KFin Technologies Ltd. with SOA University, covering transaction systems, regulatory compliance, data analytics, and cloud/DevOps.
**Generative AI Systems:** Certified by Microsoft via LinkedIn Learning.
**Advanced Problem Solving:** Certified by HackerRank for demonstrating strong proficiency in Data Structures and Algorithms.
**LeetCode 500 Days Badge / GitHub Traction:** 500+ day problem-solving streak and active open-source contributions.
**Full Stack Web Development (100xDevs):** Cohort-based training by Harkirat Singh on scalable systems and backend design.